

ISE-A1 • Problem and Stakeholder Evidence Brief

This brief argues one thing: checking a postgraduate course timetable by eye goes wrong easily, and it is worth spending one semester on a small tool.

What this project will do: take a course-registration plan and the credit constraints of the programme requirements as input, decide conflicts automatically, and output a conflict-free timetable suggestion plus a justified order to register in, with a target of finishing **within 5 minutes**.

0. Background

I am a 2026-entry master's student at UCAS (085402 Communications Engineering), and in my first year I take centralised teaching courses at the Yuquan Road campus. The university rules that students in centralised teaching cannot register for courses across campuses, and at the same time they must meet the total-credit and degree-course-count requirements of the programme requirements.

Planning one semester's courses means doing three things at once: picking candidate courses out of the official course-planning workbook issued by the academic affairs office; copying down " weekday + period + teaching week" for each course from the public course listing (jwba); and then arranging them into a weekly timetable to see whether anything clashes. The hard parts are: **the course-planning workbook has no times and no teaching weeks**; " does this count as a clash" needs three conditions to hold at once; and many required courses run in the autumn semester only, and once a section fills up you have to switch sections —**and the whole time skeleton of the timetable is rearranged with it**. Checking it by hand takes about 30 minutes, and **both the version I arranged by hand and the version the AI arranged went wrong**.

The flow I will build:

1. First read the programme requirements, and fix how many credits this semester has to cover and which degree courses must be taken;
2. Copy the candidate courses out of the course-planning workbook into a table of my own—but that table **has no teaching weeks and no weekday/period**, so the times have to come from another step;
3. For each course, go to the " time and place" page of the public course listing (jwba) and note down the **weekday + period + teaching week**;
4. Put these into the weekly timetable and check cell by cell whether anything clashes—deciding " does this count as a clash" must look at weekday, period and teaching week **at the same time**, and note which courses run in only one semester (miss it and you wait a year);
5. Register in order inside the window; during registration you may hit " the section I

want is already full” and have to switch sections —**once a section changes, every arrangement for the course changes;**

6. For anything not obtained, or where you were forced to switch sections, go back to step 4 and re-check the whole table.

What happens if this goes badly: if the semester is not planned properly, the shortfall has to be made up in later semesters. So once something goes wrong inside the window (section full, autumn-only), it is not ” one course fewer” —it is that **the timing of the whole timetable has to be rearranged.**

Terminology

- **Centralised teaching** = taking all courses on campus in the first year of the master's; **Yuquan Road** is one of the campuses (the others are Yanqi Lake and Zhongguancun).
- **Which courses I can take:** students in centralised teaching **may not register for courses across campuses. Source:** item 2 of ” VIII. Points to note” in the university's 《2026 年秋季学期选课相关通知》(” Notice on Course Registration for the 2026 Autumn Semester”), original text —” Students in centralised teaching may not register for courses across the Yanqi Lake campus and the urban campuses. For the campus where a course is taught, check the timetable in the 'registration system', or use the 18th digit of the course code: 'H' is the Yanqi Lake campus, 'Y' is the Yuquan Road campus and 'Z' is the Zhongguancun campus” (translated from the Chinese original).
- **The official course-planning workbook** (S1) = the Excel sent out by the academic affairs office: it states **which courses will run in this academic year and in which semester**, but it has **no teaching weeks and no weekday/period**.
- **The public course listing, jwba** (S2) = the site jwbaucas.ac.cn; every course has a ” time and place” page: **it has the teaching weeks, the weekday and the period, and the enrolment limit / the number already enrolled.**
- **The programme requirements** (S5) = the university document 校发培养字(2025)92 号: it sets the total credits and the **minimum number of courses and minimum credits for specialism degree courses.**
- **Registration rules: first come, first served**, submitted inside a fixed window; a course often has **several sections** (parallel sections run at different times or by different teachers, distinguished officially by the code suffix -01, -02..., also called ” class 1, class 2” in speech); some courses run **in the autumn only**, and missing them means waiting until next year.
- **Terms:** **period** = which class of the day it is; **teaching week** = which week of the semester it is; **conflict** = the same weekday, overlapping period ranges and overlapping teaching weeks, all three **at the same time**; **frozen / snapshot** = saving the data that was crawled into a file that is no longer modified, and recording the capture time and the hash with it.
- **Abbreviations:** S1 - S6 = the six sources in the source table in Section 1; E1 - E3 = the three pieces of evidence in the ledger in Section 1.2; SEP = the UCAS

postgraduate management system; below, ” 自然辩证法概论” (Dialectics of Nature, 180213010108MB001Y) is shortened to **Dialectics of Nature**, ” 新时代中国特色社会主义思想理论与实践” (Theory and Practice of Socialism with Chinese Characteristics for a New Era, 180213030500MB001Y) to **the New Era course**, and the two parallel sections of ” 学术道德与学术写作规范” (Academic Ethics and Academic Writing, 180093081000PB001Y) to **Sub-track / General track**.

1. Evidence ledger

1.1 Sources

Source	Type	How it was obtained	When
S1 The Excel 《开课计划表》 (course-planning workbook) and 《核心课和专业课列表》 (list of core and specialism courses) issued by the academic affairs office	Official document	Issued by the university	2026-08-28 version
S2 The public course listing jwbaucas.ac.cn (list pages + each course's ” time and place” page)	Official online system	Crawled and frozen by me using coding tools	2026-09-27 version
S3 The course list I produced by hand (two versions: 9-05, 9-15)	My own manual output	Exported and saved by me manually	9-05 / 9-15
S4 The course list generated by an AI Agent (Hermes) from source S1	AI output	The AI crawled the official ” time and place” pages and arranged it, and I then checked every row	2026-08-28
S5 The programme requirements document (校发培养字 (2025) 92 号)	Official document	Issued by the university	—
S6 The university's 《2026 年秋季学期选课相关通知》 (” Notice on Course Registration for the 2026 Autumn Semester”)	Official document	Issued by the university	Autumn 2026

Sources fall into two types: ① **officially published** —S1 (course-planning workbook), S2 (public course listing, crawled and frozen by me on 2026-09-27), S5 and S6 (programme requirements, registration notice); ② **the course list I produced by hand** —S3 (two registration plans: 9-05, 9-15). **S4 does not count as a separate source:** it is an artefact

the AI derived from S1, and it is only used in Section 5 as a case where ” the AI took part in arranging the timetable and got it wrong” .

Also, ” **the course-library snapshot**” comes up repeatedly below: it is a snapshot of course data **crawled from the academic system** (with fields such as teaching week, weekday/period, enrolment limit and number enrolled). **It is not a source this project uses** (this project only uses the data of S1 and S2 and the rules of S5 and S6); the AI's version of the timetable merely referred to it —and its teaching-week/time fields disagree with the official pages. **When it came to actually registering, both of my own versions and the AI's version went wrong** (see 2.1).

1.2 Evidence ledger

ID	Title	Source	Content
E1	The manual version and the AI version each went wrong	S3 + S4	① Of the 14 courses in the manual version, 3 are placed in the same slot (Computer Algebra, Digital Image Processing and Analysis, and Deep Learning Methods and Applications are all Thursday, periods 10-12), so the three are mutually exclusive and at most one of them can be taken; ② while laying out the plan the AI version referred to teaching weeks from the course-library snapshot, and the AI's understanding of teaching weeks is wrong (Marine Acoustics, ” every Sunday” , is in fact week 3 only ; the Saturday classes of Pattern Recognition and Dialectics of Nature are in fact week 6 only), which led it to give a wrong registration recommendation.

ID	Title	Source	Content
E2	During registration a required course had to switch sections, and the whole timetable was rearranged	S2 + S3	Dialectics of Nature, Academic Ethics and Academic Writing — Sub-track and the New Era course run in the autumn only, and the section wanted in Dialectics of Nature was already full; the actual outcome: Dialectics of Nature moved from Tuesday period 1 + Sunday period 1 to Wednesday period 5 + Saturday period 5 ; the New Era course moved from Tuesday period 1 to Wednesday period 1 ; the General track section originally on Wednesday/ Saturday period 7 was replaced.
E3	The two timetables cover different courses	S1 + S2	One comes from the course-planning workbook (S1) and one from the public course listing (S2): after aligning them to the same semester (autumn), the public course listing has 282 course codes and the course-planning workbook has 234 courses marked " will run in autumn" , with only 223 overlapping —the public course listing has 59 extra (not in the course-planning workbook at all, including the English A/B sections) and the course-planning workbook has 11 extra. Counting rule: both files are de-duplicated by course code with the section suffix (e.g. -01) removed, and only autumn is compared.

Verification steps for each piece of evidence:

ID	Corresponding verification steps	Result
E1	Lay the manual version and the AI version out cell by cell to see whether any courses land on the same "weekday + period" ; then take the courses in doubt back to the official "time and place" page and check their teaching weeks one by one	The manual version really does have 3 courses on Thursday periods 10-12; the teaching weeks the AI version relied on disagree with the official pages (Marine Acoustics week 3 only; Pattern Recognition and Dialectics of Nature week 6 only)
E2	① Look up the "semester taught" of these required courses in the course-planning workbook to confirm whether they are autumn-only; ② take the cell-by-cell difference of the 9-05 and 9-15 timetables (course + weekday + period) to see which courses the changes land on; ③ then use the capacity field of the public course listing to check how many places are left in the remaining sections of these courses	① Dialectics of Nature, the New Era course and the Sub-track are all autumn-only in the course-planning workbook; ② the changes land exactly on these required courses (Dialectics of Nature, the New Era course, the General track); ③ all 4 sections of Dialectics of Nature are full or over capacity (422/420, 417/420, 421/420, 418/420), and the Sub-track has only section 3 left, at 44/170
E3	Align the two data sets S1 and S2 to the same semester (autumn) and re-run	Intersection 223 courses , as expected

2. Problem / value statement

2.1 Problems in the workflow

With the same set of courses on the same campus, I arranged one version each in two independent ways (a manual version and an AI version), and **both exposed problems**:

	Manual version (S3, arranged by me by hand)	AI version (S4, arranged from the official timetable)
Data it relied on	The course-library snapshot (its time fields were measured to disagree with the official pages)	The official academic "time and place" pages (the AI referred to the course-library snapshot's teaching weeks part-way through)
Size	14 courses	10 courses
Conclusion when first produced	Looked workable	Zero conflicts after checking against official data

	Manual version (S3, arranged by me by hand)	AI version (S4, arranged from the official timetable)
Pitfalls hit	1. 3 courses are placed in the same slot (Thursday periods 10-12), mutually exclusive —found only after the plan was finished. 2. Adjusting course clashes by hand costs a great deal of effort	While laying out the plan the AI used the teaching weeks from the course-library snapshot to make its registration recommendation, and its understanding of those courses was itself wrong: Marine Acoustics, " every Sunday" , is in fact week 3 only; the Saturday classes of Pattern Recognition and Dialectics of Nature are in fact week 6 only. The AI did not consider how courses are arranged across different teaching weeks and applied a one-vote veto (treating every week as if the course met at that time)
Lesson	Arranging a timetable by hand relies on the eye, and a clash is not necessarily visible	Whatever the AI returns must be checked line by line against the original source —its mistake is not miscalculation, it is that its picture of how the courses are arranged may simply be wrong

Registration of the required courses also went wrong. I planned to register the required courses in an orderly way, and only found out when I entered the system:

- **Problem 1:** I had not fixed the order to register in advance, and by the time I came to register for the important courses the section I wanted was already full. All 4 sections of Dialectics of Nature were full or even over capacity; of the 8 sections of Theory and Practice of Socialism with Chinese Characteristics for a New Era, 4 were full at 265/265 (-01 was 263/265, -06/-07 were 169/170, and only -08 had 42 places left); for Academic Ethics and Academic Writing —Sub-track only section 3 was left. Because these courses run in the autumn only, they cannot be left to the spring to make up, and missing them means waiting a year.
- **Problem 2:** the important courses have to be taken, so I could only switch to sections that still had places, and once a section changes the arrangement of the whole timetable changes. I went looking for sections with places left, so: Dialectics of Nature moved from the planned **Tuesday period 1 (+ Sunday period 1)** to **Wednesday period 5 (+ Saturday period 5)**, the New Era course from **Tuesday**

period 1 to **Wednesday period 1**, and the section of Academic Ethics and Academic Writing —General track originally on **Wednesday/Saturday period 7** was replaced.

- **Problem 3:** arranging a timetable by hand relies on the eye, and a clash cannot be seen. And once a required course moves, the whole time skeleton of the timetable changes, so time has to be spent arranging it all over again.

2.2 Value statement

The value of this project is that it turns " a person has to arrange a semester's timetable with their eyes across several sources" into " **one course-registration plan as input plus the credit constraints of the programme requirements → decide conflicts automatically → one conflict-free timetable + one justified order to register in**". Specifically:

1. **No more deciding conflicts by eye** —the decision rule is hard-coded as three conditions holding at once; and every " this course is rejected" states clearly **which weeks overlap**.
2. **The order to register in has a basis** —" how many seats are left" and " does it offered in only one semester" are combined into a single order, rather than going on feel.
3. **Changing a plan means re-running, not re-checking the whole table** —this cuts my workload enormously.

Measurable: baseline: **about 30 minutes per round as measured by me** (changing the registration by hand and then checking one plan completely through) → target: **course planning time < 5 minutes**; the designed timetable has no conflicts, and it can suggest priorities in the order to register in.

Who benefits: mainly me, twice per academic year; classmates on the same campus under the same rules can reuse the same project.

Why this is worth a semester of software work rather than a cleverer spreadsheet: the data sources and the current state of course enrolment **change every year**, and registration costs a great deal of time and effort over and over each semester, while doing it purely through an Excel spreadsheet still takes a very long time;

2.3 Quantified pain points, baseline, value measure and cost of failure

Item	Content	How it was measured
Pain point 1: the data is scattered, and the two official systems do not cover the same set of courses	Of the 732 Yuquan Road records in the course-planning workbook, teaching weeks 0 records and weekday/period 0 records ; after aligning to the same semester, the 282 codes in the public course listing for autumn and the 234 courses " will run in autumn" in the course-planning workbook overlap in only 223 —the public course listing has 59 extra and the course-planning workbook 11 extra	Joined and counted over the frozen snapshots
Pain point 2: conflict detection is three-dimensional, and doing it by hand misjudges easily	In the AI's timetable: weeks 11-18 are the New Era course, weeks 2-5 and 7-10 are Dialectics of Nature — comparing " weekday + period" alone reports a false conflict ; only by counting the teaching weeks in does it become clear there is no conflict	Read field by field from the official time-and-place pages
Pain point 3: required courses run in the autumn only, and registration pressure is heavy	All 4 sections of Dialectics of Nature are full or over capacity (422/420, 417/420, 421/420, 418/420); of the 8 sections of the New Era course 4 are full at 265/265 (another is 263/265, two are 169/170, and only 1 section has 42 places left); the Sub-track runs in the autumn only and section 1 is full at 170/170	The capacity field of the public course listing (autumn 2026)

Item	Content	How it was measured
Pain point 4: changing a plan = re-checking the whole table	If a course cannot be obtained, the remaining registration arrangements are disrupted and other courses have to be found to make up the credit requirement.	—

- **Baseline: about 30 minutes per round as measured** (changing the registration by hand and then checking one plan completely through). How it was measured: timed with a stopwatch
- **Value measure: target < 5 minutes**, with a conflict-free course arrangement produced. How it was measured: same input, same flow
- **Cost of failure:** ① conflicts missed → rearrange; ② not enough credits → does not match the programme requirements; ③ the course is already full → that course cannot be taken, only a different section; ④ **has actually happened (E2):** required-course sections all full + autumn-only, forced to switch sections, **whole timetable rearranged**. How it was measured: ①②③ state outcomes only, no invented probabilities; ④ has measured data

2.4 Comparison with other options (non-software options + off-the-shelf tools)

Option	How it is done	Cost	Why it is not enough (a reason is required)
A. An Excel template + a printed timetable	Fixed headers, filled in and compared by hand	3-4 hours per semester	The rule of judging three conditions at once lives only in my head; the courses on offer change every year, and the plan has to be redone against the credits already taken

Option	How it is done	Cost	Why it is not enough (a reason is required)
B. Find an off-the-shelf tool / site	Search GitHub for similar projects and Bing for keywords, then look at each one's features and data	Half a day	Such tools really do exist: HMBLankcat/UCAS-course-planner already does —campus filtering by the 18th digit of the code, week-level conflict detection, capacity and number enrolled, and degree-course credits with the "2+2" structure check ("2+2" = 2 core courses of the first-level discipline + 2 specialism courses). But it cannot solve the three things needed here: ① the data is "the project's built-in autumn course data", with no semester field → "can this course wait until spring" cannot be answered; ② there are only conflict hints and filtering, there is no order to register in (there is no sorting / remaining-places / ordering logic in the source); ③ the data is a self-supplied static JSON, with no source and no capture time, and the coverage gap between the two official sources (59 / 11 courses) is not handled. There is also luyuzhe-1/ucasnanjing-course-planner (5 stars on GitHub) and GOODLIZI/visualized-course-planner-for-UCAS-students (1 star), both of the same "single-semester static data" type. ④ It

Scope note: the options compared in this section, and the project to be followed below, all use only the **current season's** official course data, and **do not include past semesters' registration records and course records**: ① past offerings and enrolments cannot tell you which sections will fill this season (courses, sections and teachers change every year, and it is common for the same course to have a different set of sections); ② in practice only about **20%** of courses have two or more years of history with enrolment numbers, which is too thin a sample, and using it for ordering would be less reliable, not more;

This project: one course-registration plan as input plus the credit constraints of the programme requirements → decide conflicts automatically → output a conflict-free timetable + a justified order to register in. Neither of the two routes A and B above is enough to solve the problems in 2.1, so this is the route to take. One difference from B is that it is **one-stop:** " **credits already earned / clauses of the programme requirements already met**" is taken as a **one-off summary input** (copied by hand from SEP, without bringing in per-course historical registration records), and the output states both " which courses to take this season" and " what is still missing" .

2.5 Boundaries for this semester: scope, non-goals, assumptions, stop conditions

- **Scope:** one course-registration plan as input plus the credit constraints of the programme requirements → decide " weekday × period × teaching week" conflicts automatically → output a conflict-free timetable + a justified order to register in (both official inputs in the input are frozen and archived first). On top of that: every rejection states clearly **which weeks overlap**; every course is labelled with its **semester taught**; and courses that run **in a single semester only and whose sections are already full** are listed separately.
- **Non-goals:** **no automatic registration**, this is registration advice only.
- **Assumptions:** the sources can still be obtained (so a frozen snapshot with a hash can be produced); teaching weeks are integer ranges and may have gaps; " courses that must be taken" are given by a person, not inferred by the tool.
- **Stop conditions:** if the semester of a course cannot be obtained from any source and cannot be recovered, **stop and report, do not assume it can be postponed**; if the public sources one day stop publishing teaching weeks/periods, narrow the scope to a " timetable consistency checker" rather than plugging the gap with guessed data.

3. Stakeholders / workflow

3.1 The five roles

Role	Who it is	Concrete impact
User	Me	Go through this whole flow once every semester

Role	Who it is	Concrete impact
Affected party	Classmates on the same campus under the same rules; my supervisor	Where the courses are placed directly determines how my research time is cut up
Decision-maker	The teaching departments and the academic affairs office	They publish the plans and set the rules
Constraint	Yuquan Road only (18th digit of the code = Y); no cross-campus registration; a minimum credit for degree courses; courses are first come, first served;	Determines which courses I can actually put into the registration table and which courses I should register for first
Current responsibility	Everything is decided and checked by me alone, by hand, with no tool and no way to share the load with anyone else	Costs a great deal of time and effort

3.2 Workflow

Step	What is done	Dependency
1	Read the programme requirements and confirm how many credits must be earned	Not in any data source —manual only
2	Assemble the candidate courses + their times	Two official sources joined by hand (E3)
3	Collect capacity / number enrolled for each course	Only S2 and S3 have it
4	Check course arrangement conflicts	Three conditions must be judged at once (E1, E3)
5	Decide the order to register in	A completely different ordering problem (E2)
6	Submit inside the window	First come, first served; failure means rearranging (E2)
7	Rearrange what was not obtained	Go back to step 4 and re-check the whole table (E1)

4. Normal / boundary / misuse scenarios

4.1 Normal scenario: what it looks like when the system runs

Input: one course-registration plan + the credit constraints of the programme requirements —where the registration plan is assembled from two **official frozen snapshots** (course-

planning workbook, 2026-08-28 version + public course listing, 2026-09-27 version), plus the "courses that must be taken" that I designate myself. Every run uses the same input, so the result is reproducible and traceable.

The five steps and outputs of one normal run:

1. **Assemble the input:** join the two sources by course code into the Yuquan Road subset (18th digit = Y; **732 raw records** for Yuquan Road in the course-planning workbook → **413 course codes** after de-duplication), then filter out by the programme requirements anything whose credits do not match, giving the candidate pool;
2. **Fill in the times:** for each course take **weekday + period + teaching week** from the "time and place" page of the public course listing (these two items are empty in the course-planning workbook, so this is the only way to fill them);
3. **Decide conflicts:** **same weekday $\cap$ period ranges intersect $\cap$ teaching weeks intersect**, all three holding at once counts as a conflict. For example, given the 9-15 plan as input, the system reports that Computer Algebra, Digital Image Processing and Analysis, and Deep Learning Methods and Applications **all collide on Thursday periods 10 - 12** and are **mutually exclusive, at most one can stay** —and states that the overlapping weeks are weeks 2 - 3 and weeks 6 - 16 (Deep Learning has no class in weeks 4-5, so it is not a continuous overlap across the whole semester);
4. **Order the registration:** sort on two dimensions, "**is it offered in only one semester**" and "**how many places are left in the section**". For example section 1 of the Subtract is full at 170/170 and section 3 has 44/170 left, so section 3 goes first; autumn-only courses always rank ahead of other courses of the same period;
5. **Output:** one **conflict-free timetable** + one **separate order to register in**; every "this course is rejected" states **which weeks overlap**; every course is labelled with its **semester taught**.

Criteria for a normal run: re-running the same input gives the same conclusion (reproducible); every number can be traced back to its source and capture time (traceable); this round's time target is **< 5 minutes**, against about **30 minutes** for one manual round.

4.2 Boundary scenario:

- **Courses that neither timetable covers:** the public course listing has 59 extra and the course-planning workbook 11 extra, and **neither can be checked**. The system can only mark "**needs manual confirmation**".
- **Key fields in the course-planning workbook are empty:** teaching weeks and weekday/period are **0 records each**, and "enrolment limit" may also be **blank**. A blank can only be taken as "**unknown**", not as "unlimited" (only values shown as / are the official "no enrolment limit"); the time has to come from the other source, otherwise that single course cannot be checked at all.
- **The official numbers themselves go out of bounds:** values such as **422/420**, where the number enrolled exceeds the limit, do occur. The system can neither count it as "

places still available” nor throw it away as dirty data —**it is precisely what shows that this course must be registered for**, and has to be handled separately.

- **Snapshots go stale**: the enrolled numbers crawled on 9-05 and those crawled on 9-15 are different (10 days apart). The system **cannot know by itself that the data is stale**, only by stating the **capture time** in the output; ordering from an old snapshot gets the order wrong.
- **Half-semester courses squeezed into the same cell**: weeks 11-18 are the New Era course, weeks 2-5 and 7-10 are Dialectics of Nature —they occupy the same cell in the weekly table and yet **do not count as a conflict**. This is exactly the easiest place to get it wrong: comparing ” weekday + period” alone produces a **false positive**, and ignoring the teaching weeks produces a **miss**; and the teaching weeks themselves have gaps (such as 2-5, 7-17), so the rule has to be able to take in these discontinuous ranges. The system must report **the specific weeks that overlap**, otherwise it is impossible to judge whether it is acceptable.
- **Boundary of the code range**: courses whose 18th digit is H (Yanqi Lake) or Z (Zhongguancun) are **directly excluded** for me, which is a different matter from a ” time conflict” ; the two must be reported separately, with a reason for each.

4.3 Misuse scenarios: who might use it that way, what the consequences are and how to block it

- **Turning it into an automatic registration program**: it would break the fairness of first come, first served. The way to block it is to hard-code the boundary —the system only outputs advice and an order and **does not submit on anyone's behalf**.
- **Reading other people's registration records**: it would invade classmates' privacy. Only public sources are used (course-planning workbook + public course listing), and no personal registration data is touched.
- **Letting the AI invent section data and make a plan ” look workable”** : this is exactly the pitfall I fell into this time (the AI version relied on wrong teaching weeks). Every section must be traceable to a **source + capture time**; if it does not match, raise an error and do not fill in a ” plausible-looking” time.
- **Treating an old snapshot as ” current”** : it would treat an already-full section as having places left. The time point must be stated in the output, and a stale one must be re-crawled.
- **Treating a blank enrolment number as ” unlimited”** : the official ” enrolment limit” **blank = unknown** (only values shown as / are ” no enrolment limit”). Every blank is treated as ” unknown” , never as ” choose freely” .

5. AI use and source-verification log

AI tools and tasks used: the Hermes Agent (an AI coding assistant I run locally, with a command-line interface). Used for: crawling and parsing the official pages, writing check scripts, joining the data, generating charts and documents, and helping me sort out the

argument structure while writing.

Where it said something wrong, and how I found out:

#	The AI tool's conclusion	My verification method	Conclusion
1	The AI's understanding of course teaching weeks was wrong: for example it held that "Marine Acoustics, Sunday periods 5-7. Every week."	The official "time and place" detail page	Overtured: Sunday has one class only, in week 3
2	"Teaching weeks and weekday/period are only available in the course-library snapshot"	The public course listing's list page + detail pages	Overtured: the official detail pages carry both items
3	AI script error: "the enrolment limit is not a number"	Went back to all the raw values of that column to look at their shape	Overtured: / is the official way of writing "no enrolment limit" —the AI read "unlimited" as "missing data"
4	AI script flagged: "discontinuous teaching weeks = data anomaly"	Counted the distribution over the whole set	Overtured: 237 of 561 course records (42%) have discontinuous teaching weeks —courses simply do not meet every week, and this is normal

#	The AI tool's conclusion	My verification method	Conclusion
5	My premise for building this tool was that " no off-the-shelf tool exists " (" I searched and did not find one")	Three rounds of GitHub API searches (course timetable clash checker, timetable conflict detection, ucas course planner) + Bing keyword searches, then read the source code and data of the closest repository	Overtuned: found 3 UCAS course planners, and the closest, HMBblankcat/UCAS-course-planner, already implements campus filtering by the 18th digit of the code, week-level conflict detection , capacity and number enrolled, and degree-course credits with the " 2+2" structure check. The original claim was changed to: tools do exist, but they do not cover the three things needed here —① semester taught (the data is built-in autumn single-semester data with no semester field) ② the order to register in (there is no sorting / remaining-places / ordering logic in the source) ③ source and time point (self-supplied static JSON, and it does not handle the 59 / 11 coverage gap between the two official sources).

What advice I refused from it: ① making " automatic registration" a feature (it breaks fairness); ② using the course-library snapshot directly as the conflict-checking input and no longer checking against the public course listing (for the reasons in rows 1 and 2 of the table above); ③ using " historical course data" to assess which courses to register for and set priorities (because offerings change every year, and only about 20% of courses have two or more years of history with enrolment numbers, which is unconvincing).